Linear Systems Theory

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Module 3

Lecture 3

Math Preliminaries: Linear Algebra

Diagonalisation

Diagonal Matrices and Properties

A square matrix with non-diagonal entries being zeros

$$a_{ij} = 0 \;\forall\; i \neq j$$

Properties of diagonal matrices:

- Rank of a diagonal matrix is equal to its non-zero diagonal entries
- The diagonal elements of a diagonal matrix are its Eigen values
- Product of two diagonal matrices $\mathbf{A}$ and $\mathbf{B}$ is another diagonal matrix and is also commutative : $\mathbf{AB} = \mathbf{BA}$
- The n^{th} power of a diagonal matrix is also diagonal and $[\mathbf{A}^n]_{ii} = a_{ii}^n$
- The inverse of a diagonal matrix with non-zero diagonal entries is also diagonal and its diagonal elements are:

$$[\mathbf{A}^{-1}]_{ii} = \frac{1}{a_{ii}}$$

Matrix Diagonalization

If a square matrix $\mathbf{A}$ is similar to a diagonal matrix $\mathbf{D}$, then it is said to be diagonizable

- Alternately, for a linear transformation $f : V \to V$ represented by $\mathbf{A}$, there exists a basis for V which results in a diagonal matrix $\mathbf{D}$ corresponding to f
- That basis for V is given by the eigen vectors of $\mathbf{A}$
- Let $\lambda_1, \ldots, \lambda_n$ be the eigen values and $\mathbf{e}_1, \ldots, \mathbf{e}_n$ be the eigen vectors of an $n \times n$ square matrix $\mathbf{A}$ which is diagonalizable
- Then $\mathbf{e}_1, \ldots, \mathbf{e}_n$ form a basis for $\mathbb{R}^n$
- Let $\mathbf{D} = \mathbf{diag}\begin{bmatrix}\lambda_1 & \ldots & \lambda_n\end{bmatrix}$ and $\mathbf{E}$ be a matrix with eigen vectors $\mathbf{e}_1, \ldots, \mathbf{e}_n$ as its columns, then

$$\mathbf{AE} = \mathbf{ED} \implies \mathbf{D} = \mathbf{E}^{-1}\mathbf{AE}$$

- **Theorem:** A $n \times n$ square matrix is diagonalizable if and only if it has n linearly independent eigen vectors

Matrix Diagonalization: Example

Eg. Diagonalise the matrix $\mathbf{A} = \begin{bmatrix} 2 & 6 \\ 0 & -1 \end{bmatrix}$

Solution: $\lambda = -1, 2$ and the eigen vectors are $\begin{bmatrix} -2 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

Exercise 1

Diagonalise the matrix $\mathbf{A} = \begin{bmatrix} 6 & 2 \\ -1 & 3 \end{bmatrix}$

Defective Matrices and Generalised Eigen Vectors

Defective Matrices

Matrices with eigen values whose algebraic multiplicity exceeds the geometric multiplicity are **defective**.

- Eg. $\mathbf{A} = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$; $\lambda = 3, 3$ For $\lambda = 3$, algebraic multiplicity is 2 and geometric multiplicity is 1
- Defective matrices of size $n \times n$ have less than n linearly independent eigen vectors
- Therefore defective matrices are not diagonalizable
- Then, how to get a basis for $\mathbb{R}^n$ comprising of eigen vectors of $\mathbf{A}$?
  - **Generalised eigen vectors**

Generalised Eigen Vector

A generalised eigen vector of an $n \times n$ matrix $\mathbf{A}$ corresponding to the eigen value λ is a non-zero vector $\mathbf{x}$ satisfying:

$$(\mathbf{A} - \lambda \mathbf{I}_n)^p \mathbf{x} = \mathbf{0} \tag{1}$$

where p is a positive integer. Equivalently, it is a non-zero element of the nullspace of $(\mathbf{A} - \lambda \mathbf{I}_n)^p$.

- The value of p corresponding to an eigen value λ can be atmost the algebraic multiplicity of λ denoted by k_λ
- Eigen vectors are generalised eigen vectors with $p = 1$
- Generalised eigen vectors form an eigen vector basis for $\mathbb{R}^n$ in case the matrix is defective

Finding Generalised Eigen Vectors

For an $n \times n$ matrix, let the algebraic multiplicity of an eigen value λ be k_λ.

$$(\mathrm{A} - \lambda \mathbf{I}_n)\mathbf{x}_p = \mathbf{x}_{p-1} \ \forall \ p = 2, \ldots, k_\lambda \tag{2}$$

- Let $\mathbf{x}_1$ by the eigen vector corresponding to $p = 1$, that is, $(A - \lambda \mathbf{I}_n)\mathbf{x}_1 = \mathbf{0}$
- As per Eq.2,

$$(A - \lambda \mathbf{I}_n)\mathbf{x}_2 = \mathbf{x}_1$$
$$(A - \lambda \mathbf{I}_n)^2\mathbf{x}_2 = (A - \lambda \mathbf{I}_n)\mathbf{x}_1 = \mathbf{0}$$

Therefore, solving Eq.2 is equivalent to finding $\mathbf{x}$ in Eq.1, for different values of p

- In this manner, generalised eigen vectors form a chain:

$$\mathbf{x}_p \xrightarrow{\mathrm{A}-\lambda\mathrm{I}} \mathbf{x}_{p-1} \xrightarrow{\mathrm{A}-\lambda\mathrm{I}} \ldots \xrightarrow{\mathrm{A}-\lambda\mathrm{I}} \mathbf{x}_2 \xrightarrow{\mathrm{A}-\lambda\mathrm{I}} \mathbf{x}_1 \xrightarrow{\mathrm{A}-\lambda\mathrm{I}} \mathbf{0}$$

Generalised Eigen Vector: Example

- Eg. $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. Find a basis for $\mathbb{R}^2$ comprising of generalised eigen vectors of $\mathbf{A}$.

  Solution: $\lambda = 1, 1$

Block Matrices and Jordan Normal Forms

Block Matrix

A block matrix is a matrix defined using smaller matrices or submatrices called blocks.
Egs:

$$X_{4\times4} = \begin{bmatrix} A_{2\times2} & B_{2\times2} \\ C_{2\times2} & D_{2\times2} \end{bmatrix} \; ; \; X_{4\times4} = \begin{bmatrix} A_{1\times1} & B_{1\times3} \\ C_{3\times1} & D_{3\times3} \end{bmatrix}$$

where $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$ are submatrices and referred to as blocks of $\mathbf{X}$

- The submatrices can be of any dimension as long as they are together in consistence with the block matrix
- The blocks along the diagonal are referred to as diagonal blocks
- Two block matrices of size and sqaure diagonal blocks multiply similar to matrix multiplication

$$\begin{bmatrix} A_1 & B_1 \\ C_1 & D_1 \end{bmatrix} \begin{bmatrix} A_2 & B_2 \\ C_2 & D_2 \end{bmatrix} = \begin{bmatrix} A_1A_2 + B_1C_2 & A_1B_2 + B_1D_2 \\ C_1A_2 + D_1C_2 & C_1B_2 + D_1D_2 \end{bmatrix}$$

Block Diagonal Matrix

A block matrix which is a square matrix, and whose diagonal blocks are square matrices and off-diagonal blocks are zero matrices. Eg:

$$A = \begin{bmatrix} A_1 & 0 \\ 0 & A_2 \end{bmatrix}$$

Properties of block diagonal matrices with n diagonal blocks:

- $\det(A) = \det(A_1)\det(A_2)\ldots\det(A_n)$
- $\mathrm{tr}(A) = \mathrm{tr}(A_1) + \mathrm{tr}(A_2) + \ldots + \mathrm{tr}(A_n)$
- The inverse of a block diagonal matrix is another block diagonal matrix, composed of the inverse of each diagonal block: $A^{-1} = \mathrm{diag}(A_1^{-1}, \ldots, A_n^{-1})$
- The eigen values of a block diagonal matrix A are combination of eigen values of $A_1, \ldots, A_n$

Jordan Normal Form or Jordan Cannonical Form or Jordan Matrix

A Jordan matrix is a block diagonal matrix whose diagonal blocks are all Jordan blocks. It is a generalisation of diagonal matrices

- A Jordan block associated with λ is a square, upper triangular matrix whose entries are all λ on the diagonal, all 1 on the entries immediately above the diagonal, and 0 elsewhere.

$$\text{Eg. } J = \begin{bmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{bmatrix}$$

- In a Jordan matrix, each jordan block can be associated with a different value of λ
- Eg. of a Jordan Matrix with two Jordon blocks of size 3×3 and 1×1 corresponding to $\lambda = 2$ and $\lambda = 3$

$$J = \begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

Properties of Jordan Matrices

- Diagonal matrix is a special case of Jordan matrix with each diagonal element forming a Jordon block
- There exists a Jordan matrix that is similar to every sqaure matrix i.e., every square matrix $\mathbf{A}$ can be transformed into a Jordan matrix.

$$\mathbf{J} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$$

- The λ values of Jordan blocks in $\mathbf{J}$ are given by the eigen values of $\mathbf{A}$
- Model matrix $\mathbf{P}$ comprises of the generalised eigen vectors of $\mathbf{A}$ as its columns

Jordan Matrices: Example

Find the Jordan form of the matrix $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ -1 & 4 \end{bmatrix}$

Solution: $\lambda = 3, 3$

Symmetric Matrices

Symmetric Matrices and Properties

Symmetric matrix is a square matrix that is equal to its transpose

$$\mathbf{A} = \mathbf{A}^T$$

$$a_{ij} = a_{ji}$$

Properties of symmetric matrices:

- Every square diagonal matrix is symmetric
- The sum and difference of two symmetric matrices is symmetric
- $\mathbf{A}^n$ is symmetric, if $\mathbf{A}$ is symmetric
- If $\mathbf{A}^{-1}$ exists, it is symmetric iff $\mathbf{A}$ is symmetric
- The eigen values of a symmetric matrix are all real
- The eigen vectors corresponding to distinct eigen values of a symmetric matrix are orthogonal to each other
- Every symmetric matrix is diagonalizable i.e., its eigen vectors are linearly independent

Eigen Decomposition and Singular Value Decomposition (SVD)

Eigen Decomposition

A decomposition which decomposes a square matrix **A** into a set of its eigen values and eigen vectors

$$\mathbf{A} = \mathbf{E}\mathbf{D}\mathbf{E}^{-1}$$

where **E** is a matrix with eigen vectors of **A** as its columns and **D** is a diagonal matrix with eigen values of **A** as diagonal elements

- It is an alternate interpretation to matrix diagonalization
- Eigen decomposition is possible only if the eigen vectors of **A** are linearly independent
- This decomposition has its applications in calculating powers of a matrix and matrix exponential

$$\mathbf{A}^n = \mathbf{E}\mathbf{D}^n\mathbf{E}^{-1}$$
$$e^{\mathbf{A}} = \mathbf{E}e^{\mathbf{D}}\mathbf{E}^{-1}$$

- What if **A** is rectangular ? - **Singular Value Decomposition**

Singular Value Decomposition (SVD)

Singular Values of a Matrix

- Let $A \in \mathbb{R}^{m \times n}$; $\quad \textbf{rank}\,(A) \leq \textbf{min}(m, n)$
- $AA^T \in \mathbb{R}^{m \times m}$, is a symmetric matrix.
- Let $\lambda_1, \ldots, \lambda_r, \quad r \leq m$ denote the non zero (these are all real) eigen values of AA^T.
  $\textbf{rank}\,(A)\,A^T = r$
- The non zero singular values of A are $\sqrt{\lambda_i}, \; i = 1, ..., r$.
- The remaining singular values of A are zero.

Singular Value Decomposition (SVD)

The Singular Value decomposition is the decomposition of a matrix into three other matrices:

$$\mathbf{A}_{m\times n} = \mathbf{U}_{m\times m}\mathbf{S}_{m\times n}\mathbf{V}^T_{n\times n}$$

where **U** and **V** are orthogonal matrices and **S** is a matrix containing elements along the main diagonal

- This is a generalisation of eigen decomposition to any $m \times n$ matrix
- The diagonal elements of **S** are the singular values of **A**
- The columns of **U** are the eigen vectors of $\mathbf{A}\mathbf{A}^T$ (or) the left singular vectors of **A**
- The columns of **V** are the eigen vectors of $\mathbf{A}^T\mathbf{A}$ (or) the right singular vectors of **A**

Singular Value Decomposition (SVD)

- SVD and Eigen decomposition coincide if and only if $\mathbf{A}$ is symmetric and positive definite
- An orthogonal basis for each of the fundamental subspaces of a matrix $\mathbf{A}$ is given by Singular Value Decomposition (SVD)
  - Basis for $\mathcal{C}(\mathbf{A})$: First r columns of $\mathbf{U}$
  - Basis for $\mathcal{N}(\mathbf{A})$: Last $n - r$ columns of $\mathbf{V}$
  - Basis for $\mathcal{R}(\mathbf{A})$: First r columns of $\mathbf{V}$
  - Basis for $\mathcal{N}(\mathbf{A}^T)$: Last $m - r$ columns of $\mathbf{U}$

Overview

Summary: Mod 3 Lecture 3

- Diagonalisation
- Generalised Eigen vectors
- Jordan Matrices
- Singular Value Decomposition

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- State transition matrix
- Solutions to LTI and LTV systems